

Rishabh Verma

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EDUCATION

New Jersey Institute of Technology – Ying Wu College of Computing

Newark, NJ

Master of Science in Data Science

Jan 2026 – May 2027

- **Relevant Coursework:** DS644 Big Data Systems, DS650 Data Visualization and Interpretation, DS675 Machine Learning, MATH661 Applied Statistics, DS637 Python and Mathematics for Data Science, DS680 NLP, DS667 Deep Learning, CS683 Software Project Management

New Jersey Institute of Technology – Ying Wu College of Computing

Newark, NJ

BS in Data Science (BS/MS track), Minor in Business | GPA: 3.3/4.0

Sep 2021 – Dec 2025

- **Relevant Coursework:** CS288 Intensive Programming in Linux, CS482 Data Mining, CS370 Intro to Artificial Intelligence, CS493 Senior Capstone II, CS331 Database System Design and Management, MATH478 Statistical Methods in Data Science, MATH344 Regression Analysis, MATH337 Linear Algebra, FIN320 Finance Data Analytics, FIN310 Data-Driven Financial Modeling, FIN306 Blockchain Technology for Business, FIN315 Fundamentals of Corporate Finance, FIN218 Financial Markets and Institutions, ACCT117 Principles of Financial Accounting, ECON201 Economics, COM312 Oral Presentation

EXPERIENCE

Forward Deployed Engineer Intern — Cloud Business Division (MSP)

Jun 2026 – Aug 2026

Samsung SDS America (SDSA)

Ridgefield Park, NJ

- Led team coordination and project execution for an AI-enabled rail predictive-maintenance solution, delegating responsibilities, managing timelines and follow-ups, and resolving blockers to keep delivery on track
- Applied AI/ML expertise to equipment monitoring, failure prediction, and fault diagnostics; decoded system logic and outputs and connected technical components into a coherent end-to-end solution
- Worked with ETL pipelines, Kafka-based event streaming, and data lake / object storage (S3-compatible) components to move equipment-sensor data into the monitoring and prediction pipeline
- Explored retrieval-augmented generation (RAG) and Model Context Protocol (MCP)-based integrations while driving prompt engineering and rapid prototyping on AWS, designing, testing, and iterating on prompts to turn concepts into working prototypes
- Promoted explainable and responsible AI practices — including PII-aware data handling and data-lineage/traceability discussions — so predictions stayed understandable, trustworthy, secure, and governed rather than opaque automated decisions
- Evaluated Snowflake and Databricks as candidate platforms for scalable data engineering and model experimentation, and supported FinOps-style reviews of AWS consumption using Apptio to help inform architecture and investment decisions
- Translated technical capabilities into operational and business value — real-time alerts, maintenance planning, uptime, fewer breakdowns, safer operations, and on-time repairs — and supported the revenue model and pitches to investors, supervisors, and operators
- Created infographics and presentation material that turned technical and business information into a clear project story for stakeholders
- **Technical Environment:** AWS, RAG, MCP, ETL, Kafka, S3/MinIO, Data Lakes, Snowflake, Databricks, Apptio, FinOps, PII Governance, Data Lineage

Coordinator & CECC IT Specialist

Oct 2021 – Nov 2025

Career Development Services – NJIT

Newark, NJ

- Collaborated with 35 nonprofits on their websites, newsletters, and banners; sites include dblnj.org, estherinstitute.org, gracwestmanor.com, kab.org, and njit.edu
- Administered 8 projects in a waterfall structure using MS Project to bring efficiency into the critical path
- Coached 10+ student developers on website analysis, web development tools, client communication, and professional reporting standards, improving project delivery quality and client satisfaction
- Directed 2 teams building an Android app for Paralympic track events, along with websites, flyers, and newsletters
- Updated cecc.njit.edu with fresh projects, reducing phone call rates by 60% and increasing team productivity, which helped the department secure a better budget from the institute

Azure AI Consultant

May 2025 – Jul 2025

*Silicon Harlem**Harlem, NY*

- Engineered AI-enabled chatbots for small businesses in Harlem, NY, to boost sales, retain customers, and streamline processes
- Modeled multiple business plans for the company and pitched them to investors to fund smart-city initiatives
- Taught 20 kids how to build a simple chatbot and sell products door-to-door

Chief Executive Officer

Jan 2022 – Dec 2024

*Invasion Media**Indore, India*

- Achieved 80% ROI through strategic social media and web management initiatives
- Directed company vision, resource allocation, and financial planning with a focus on high-performance team growth in 3 months
- Spearheaded 200 stakeholder meetings to align business goals with marketing and operational strategy

Software & Product Manager

Jan 2022 – Dec 2024

*Invasion Media**Indore, India*

- Built 3 dynamic versions of the company website's front-end and back-end feature integration
- Designed NDA-protected backend software, managed a development team of 5, and led debugging and deployment
- Introduced internal systems to track productivity and eliminate redundancy across teams by 40%

Business Consultant & Operations Lead

Jan 2022 – Dec 2024

*Invasion Media**Indore, India*

- Developed custom business strategies for 50 clients to streamline reporting and enhance decision-making
- Evaluated performance metrics to identify inefficiencies and proposed solutions to boost employee output
- Collaborated with 50 clients to identify marketing gaps and provide tailored business transformation advice

Finance & HR Coordinator

Jan 2022 – Dec 2024

*Invasion Media**Indore, India*

- Administered payroll, tax filings, and subscription cost planning with zero compliance issues
- Formed contracts, led hiring operations, and managed HR-related risk and conflict resolution initiatives
- Directed HR initiatives including 55 contracts, hiring strategy, and 3 productivity audits

Project Manager

Feb 2019 – Nov 2024

*Shockdrops LLC**Livingston, NJ*

- Administered a team of 25, introduced growth tracking protocols, and ensured 90% staff retention
- Optimized 500 internal reports to monitor project timelines, client satisfaction, and team performance
- Drove cross-functional collaboration across development, finance, and hiring teams

Product Operations & Strategy Lead

Feb 2019 – Nov 2024

*Shockdrops LLC**Livingston, NJ*

- Created client-focused solutions by managing internal policies for hiring, vendor renewals, and new client onboarding
- Formed systems to identify cost-saving opportunities across membership fees, salaries, and contracts
- Operated directly with leadership to translate operational insights into strategic decisions

Data Science Intern

Jun 2021 – Jul 2021

*Intel Corporation**Newark, NJ*

- Devised and developed a machine learning application proficient in solving reCAPTCHA puzzles
- Integrated the application into proprietary software, polishing its functionality and efficiency
- Initiated internal networking, fostering skill development and enhancing communication abilities

RESEARCH

Researcher – Weather Forecasting & Climate Modeling

Sep 2024 – May 2025

Ying Wu College of Computing, NJIT

Newark, NJ

- Conducted advanced modeling with the WRF-Solar model to enhance short-term weather prediction accuracy
- Developed and tested machine learning optimization pipelines using Bayesian Optimization and Double Dueling Deep Q-Networks (D3QN)
- Explored alternative numerical methods (FDM, FEM, NWP) but confirmed superior accuracy with Bayesian-based tuning
- Authored two LaTeX-formatted research papers, one focused on Bayesian Optimization and the second on Reinforcement Learning integration
- Designed reproducible experiments on NJIT's GPU cluster and maintained logs in structured repositories

Finance Research Assistant

May 2024 – Feb 2025

Martin Tuchman School of Management, NJIT

Newark, NJ

- Co-authored (second author, 60% contribution) a decentralized finance research paper under Dr. Wang: *Do Global Uncertainty Factors Matter More to Cryptocurrency?* J. Risk Financial Manag. 2025, 18, 628 – doi.org/10.3390/jrfm18110628
- Applied the BEKK-GARCH model to analyze data from the Bloomberg Index and wrote the paper in LaTeX
- Presented the paper at SWFA 2025 in Texas
- Coached an intern for 3 months on data collection and the research-writing process

PROJECTS

ECG Data Processing with Hadoop MapReduce on AWS | *Java, HDFS, YARN, EC2*

Spring 2026

- Deployed a fully distributed 4-node Hadoop cluster (1 NameNode, 3 DataNodes) on AWS EC2 Ubuntu instances, installing and configuring Java 8 and Hadoop 3.x across all nodes
- Configured HDFS, YARN, and passwordless SSH (core-site, hdfs-site, mapred-site, yarn-site) to enable fully distributed processing and inter-node communication
- Developed a custom Java MapReduce job (Mapper, Reducer, and Driver classes) to process a 500MB+ ECG dataset of cardiac patients ingested into HDFS, spanning four diagnostic categories: Normal, Abnormal Heartbeat, Myocardial Infarction (MI), and History of MI
- Executed and monitored jobs through YARN, retrieved job output from HDFS, and interpreted the results across patient categories
- Diagnosed cluster bottlenecks, documented troubleshooting steps, and identified performance optimization strategies

Object Tracing and Classification on Video Streams | *PyTorch, OpenCV, Docker*

Sep 2024 – Dec 2024

- Designed and deployed a deep learning pipeline for object detection and classification on video data using pretrained transformer models
- Utilized Hugging Face models for fine-tuning classification layers; implemented real-time object tracing via OpenCV
- Containerized the solution using Docker to ensure modularity and reproducibility across environments
- Built 4 AI-driven automation tools and analytics workflows with Docker
- Integrated Ollama and Hugging Face to develop and deploy a Gradio application to fine-tune routes
- Formed an ETL pipeline to store raw vehicle-route data in MongoDB
- Constituted an environment using PyTorch, Qdrant, and ClearML to create the backend logic of the application
- Integrated asynchronous frame extraction, prediction, and annotation to optimize throughput and inference speed

All-in-One (AIO) Bot – Shade Softwares | *Automation Platform, Product Lifecycle*

Jan 2022 – May 2022

- Built a scalable automation platform enabling 300 users to execute multiple concurrent web requests while mitigating detection and throttling risks
- Sold subscription-based software, generating five-figure semi-annual revenue
- Led end-to-end product lifecycle, including architecture design, pricing strategy, customer onboarding, and technical support

F.I.R.S.T. Robotics Championship – Livingston High School | *Autonomous Robotics*

Oct 2019 – Feb 2020

- Engineered an autonomous robot capable of pathfinding to park itself, pick up a capstone, and place it on top of the game tower

- Won the State Championship and was awarded Best Logbook, which I had a hand in creating
- Qualified for Nationals (events canceled due to COVID)

Dynamic Routing System for School Buses – BaalVigyan 2018 | *Route Optimization* Mar 2018 – May 2018

- Innovated a dynamic routing system for school buses in India, optimizing transportation efficiency
- Implemented real-time tracking for students and parents, offering alternative boarding options for late or remote students
- Reduced student pick-up delays, minimized unnecessary stops, and improved driver–parent–student communication, cutting carbon emissions by 20%
- Awarded Best Design 2018 (High School)

City Trash Collection System – BaalVigyan 2017 | *Route Optimization Software* Mar 2017 – May 2017

- Designed route optimization software for efficient urban trash collection in India, resulting in annual savings of \$3,200 across three cities
- Mitigated littering and enhanced city aesthetics by 40%, contributing to Indore winning “the cleanest city in India” for the 8th time in a row
- Awarded Best Idea of 2017 (High School)

TECHNICAL SKILLS

Languages: Python, R, Java, C++, C, SQL, JavaScript, HTML, CSS, Bash

Big Data & Cloud: Hadoop, HDFS, MapReduce, YARN, AWS (EC2), Docker, MongoDB, Qdrant

ML & Data Science: Scikit-learn, NumPy, TensorFlow, PyTorch, Hugging Face, OpenCV, NLP, ClearML, Ollama, Gradio

Visualization & Analytics: Tableau, Power BI, Plotly, SAS, MATLAB

Tools: Git, Anaconda, VS Code, Kali Linux, MS Project, LaTeX

KEY SKILLS

Data Analysis, Data Mining, Data Cleaning & Manipulation, Predictive Modeling & Analytics, Machine Learning, Statistical Modeling, Quantitative Analysis, Data Visualization, Big Data Management, Deep Learning Frameworks, Text Analytics, Unstructured Data, Project Delivery, Process Improvement, Team Coordination & Management, Operations Management, Technical Writing, Marketing Analysis, Customer Engagement, Analytical Frameworks, Troubleshooting, Pen-Testing, Damage Control, Critical Thinking

VOLUNTEER EXPERIENCE

Assistant Teacher & Event Manager

Mar 2019 – Jan 2021

Sanskriti of NJ

Livingston, NJ

- Assistant teacher for Hindi, chess, and cooking for children ages 5–14
- Event manager for functions hosted by the organization

Senior Tech Support Volunteer

Dec 2019 – Mar 2020

LHS Techie Teen Tutor

Livingston, NJ

- Assisted senior citizens in resolving tech issues
- Conducted workshops at the Livingston Library familiarizing members with their devices

Instructor & Mentor

Oct 2019 – Jan 2021

Blissful Coding Club

Livingston, NJ

- Taught 20+ high school students HTML, CSS, Java, and Python
- Guided students in pioneering their projects for the Google competition

HONORS & AWARDS

1st Place, Business Consulting Competition (Dec 2025)

Dean's List, 5 semesters (Dec 2021 – Dec 2025)

Honors List, 2 semesters (Oct 2020 – May 2021)

Best Design 2018 – High School (BaalVigyan)

Best Innovation of 2017 (BaalVigyan)